All-root SEA branch audit

Fixed NIST/ANSI cohort | 27 October 2026

Observations only: all 21 curves pass the tested arithmetic. Historical selection remains unresolved.

Workload and controls

63 specializations = 21 curves x levels 3, 5, 13. All rational roots were tested with direct Y-coordinate and pre-Dewaghe ambiguous X-coordinate profiles. The native observer received only p, a, b and the auxiliary prime.

Measured: 39 rational modular roots, 78 kernel/profile checks, 35 Atkin specializations. Independently verified: 78 division-polynomial, Velu/isomorphism and quotient/root Fricke checks. Zero actual cohort failures or timeouts.

81 native invocations were retained: 9 initial missing-table controls, then 9 controls and 63 named cases. Three exceptional-invariant controls reached source errors and SIGABRT; none of the 21 cohort curves did. The first compile failure and both offline auditor versions are retained.

Independent polynomial arithmetic reconstructed every specialized table polynomial, enumerated the complete rational-root set with gcd(F,X^p-X), checked distinct kernels and trace relations, and verified quotients. Controls included direct finite-field point counting and explicit positive/negative order-3 kernels.

Source and inference

LiDIA commit 23dac491f4cd96eb7de648a5cf90eb9ec77653c8; version string 2.3.0+latte-patches-2019-05-02. Access labels alone were changed in an observer header; the static library was unchanged. This is a later lineage, not authenticated 1997 generator code.

All-root success leaves no omission-specific exceptional path in this build at these levels. Earlier code, other levels, optimizer settings and caller policies remain untested. It does not establish a historical exclusion criterion.

Provenance gap: the full fetched Git history starts with a 2013 import. NEWS says ECO was integrated in 2.1pre2; ChangeLog dates documentation to June 2001 and records September 2001 splitting-degree/crash fixes and a March 2004 schoofpart fix. The old faulty lines were not recovered.

Evidence: attempt-02/PROTOCOL.json, runs/case-00..20-ell-{03,05,13}, AUDIT.json; source-history/ChangeLog and git-history.txt. Complete capture and replay: evidence.tar.zst, BUNDLE.json.

Every curve, including ordinary behavior

Heatmap of exact rational modular root counts: darker cells mean more roots (scale 0 to 4). Published labels are blue; reconstructed omissions are orange. Labels describe recovered streams, not documented historical rejection events.

Field	Ordinal	Observed label	l=3	l=5	l=13
239	36	omitted	0	0	2
239	1562	omitted	0	1	2
239	2185	published	0	0	0
239	2320	published	0	1	0
239	2491	omitted	0	0	1
239	2682	omitted	0	1	1
239	2863	omitted	0	1	2
239	3048	published	0	1	0
192	139	published	0	1	2
192	2198	omitted	0	0	0
192	5228	published	0	1	2
192	5405	omitted	0	0	2
192	6013	published	0	1	0
521	1136	published	0	1	2
256	876	omitted	1	0	0
256	1842	published	1	1	2
384	2688	published	0	0	0
224	777	omitted	1	1	0
224	1485	omitted	1	0	0
224	5736	omitted	1	0	1
224	7026	published	4	0	0

Measured native root counts; independently checked against the complete linear-factor polynomial. Exact deterministic values; no sampling uncertainty estimated. Full field/model/order/seed/UID in frozen inputs and comparison.csv.

Patterns retained as leads

Post hoc observations: P-224 has four level-3 roots for its publication and one for each omission. P-192 and P-256 publications have one level-5 root, with none for their omissions. P-239 publications have no level-13 roots, while every omission has one or two.

Different fields require different choices. A rule in only these three root counts cannot classify the full sample: omitted P-192/2198 and published P-239/2185 and P-384/2688 share (0,0,0). Field-dependent rules remain possible, but this source handles all these paths successfully and does not reject on these patterns.

Historical test still required

Recover a dated pre-fix or standalone point counter, identify the actual relevant path, freeze its prediction without label-dependent changes, and check all 21 curves. No hypothetical reversal of a comparison is labeled recovered code. Inaccessible archive requests do not establish absence of older code.

Portable replay

Extract evidence.tar.zst; run python3 -B evidence/verify.py. It verifies 489 sealed files and repeats independent Python arithmetic only. Native programs launched by replay: zero. The archive includes the raw errors and initial missing-table attempt; no failed invocation was silently repeated.

Protocol SHA-256: b80560225b69b2cbb8370f01e26946164ef03b3d30055b31e50605a3566834ab. Seal and archive hashes: report.md and BUNDLE.json.

Sources

[1] github.com/passagemath/upstream-LiDIA, commit 23dac491f4cd96eb7de648a5cf90eb9ec77653c8, src/ec_order and ChangeLog.

[2] Same repository, initial import 0171e66, NEWS: integration in 2.1pre2. Full source-history receipts included.

[3] words.filippo.io/seeds-solved/ (16 October 2026), the historical selection question.

[4] Local LMFDB av_fq manifest, public.av_fq_isog, extraction ended 2026-10-19T16:45:49.243703+00:00. Genus-1 coverage q<=1024; no match for these fields.

This report changes no official hypothesis status, conductor result or verified transfer map. The missing criterion remains open.